

# Core Statistics for In Vitro Microbiome Data

---

*Eliana Ibrahimi - University of Tirana, Albania*

*Third INFOGUT Training School - September 28-30, 2026. Madrid, Spain*  
*Bioinformatics 101 for colon models experiments*



## Choosing the right test matters as much as a good study design

---

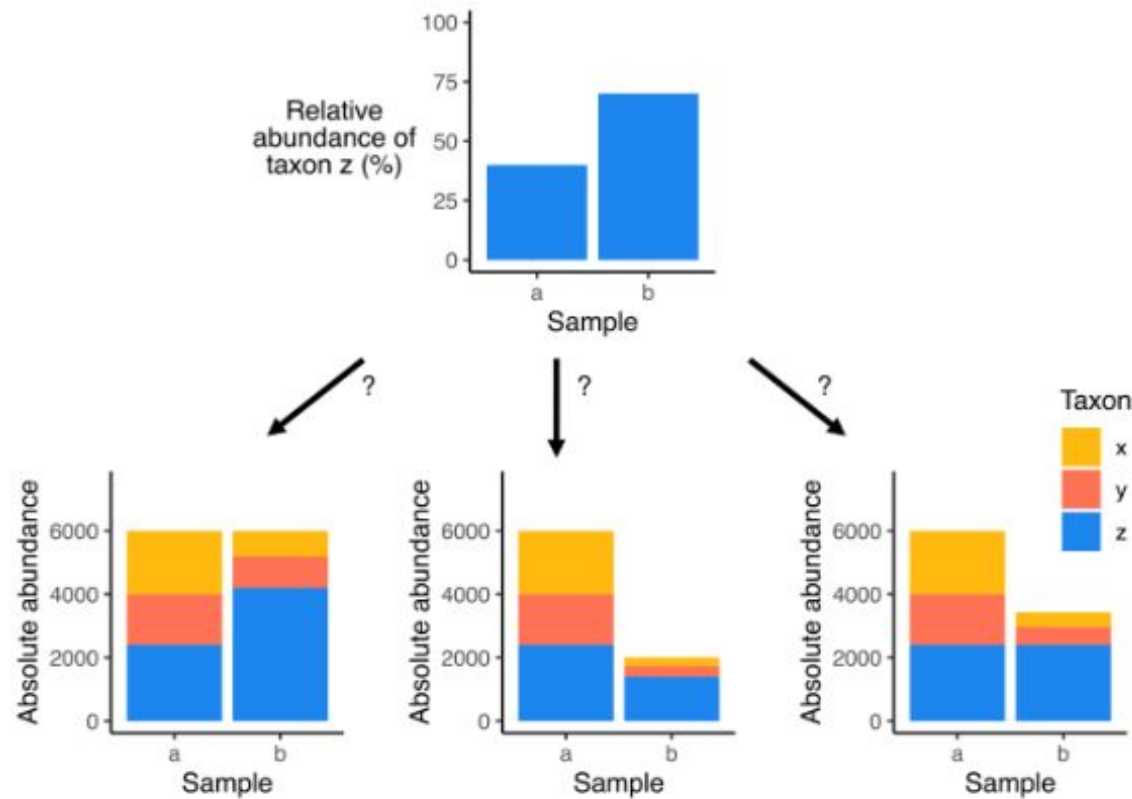
*“A clean design still gets ruined by the wrong test, or the right test run a few thousand times without correcting for it.”*

## Session overview

---

- Exploration of microbiome data
- Hypothesis testing fundamentals
- Testing diversity
- Differential abundance & compositionality
- Multiple comparisons
- Effect size vs. significance

# Relative vs absolute abundances



Douglas & Langille, 2021

- Sequencing mostly gives relative abundances that sum to a fixed total, this is called compositional data.
- If one taxon's true absolute abundance rises, every other taxon's relative share must fall to keep the total at 100%.
- Classical statistical methods treat every taxon as independent, which compositional data, by construction, is not.

Gloor et al. (2017), *Frontiers in Microbiology*

## One taxon's real change creates two fake ones

| Taxon                    | Sample 1<br>(absolute) | Sample 1<br>(relative) | Sample 2<br>(absolute) | Sample 2<br>(relative) |
|--------------------------|------------------------|------------------------|------------------------|------------------------|
| Taxon A (really doubled) | 50                     | 50.0%                  | 100                    | 66.7%                  |
| Taxon B (unchanged)      | 30                     | 30.0%                  | 30                     | 20.0%                  |
| Taxon C (unchanged)      | 20                     | 20.0%                  | 20                     | 13.3%                  |
| <b>Total</b>             | <b>100</b>             | <b>100%</b>            | <b>150</b>             | <b>100%</b>            |

*Taxon B and C's absolute abundance never changed, but their relative abundance dropped by 10 and 6.7 percentage points respectively, purely because Taxon A really doubled. A naive per-taxon test would call all three “significant.”*

## Controlling for sequencing depth

---

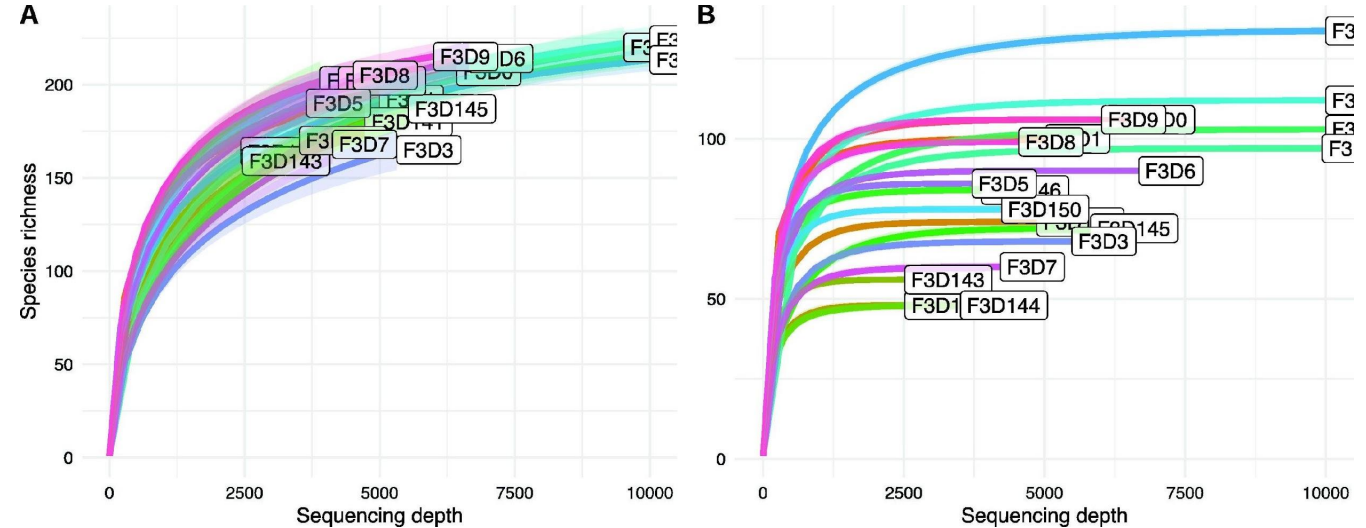
- Sequencing depths varies from sample to sample
  - Not directly related to the original amount of cells/DNA
  - More depth doesn't mean more microbes
- Greater sequencing depth  $\square$  greater taxa detection
- We need to control/normalize for sequencing depth

# Controlling for sequencing depth

## *Rarefaction*

Subsampling sequencing reads without replacement to a fixed depth so different samples have the same read count.

It's commonly used before **alpha** and **beta diversity** calculations.



Bardenhorst, et al. 2022

# Does rarefying solve the problem?

 OPEN ACCESS  PEER-REVIEWED

RESEARCH ARTICLE

## Waste Not, Want Not: Why Rarefying Microbiome Data Is Inadmissible

Paul J. McMurdie, Susan Holmes 

Published: April 3, 2014 • <https://doi.org/10.1371/journal.pcbi.1003531>

<https://doi.org/10.1371/journal.pcbi.1003531>

| Original Abundance |     |      | Rarefied Abundance |     |     |
|--------------------|-----|------|--------------------|-----|-----|
|                    | A   | B    |                    | A   | B   |
| OTU1               | 62  | 500  | OTU1               | 62  | 50  |
| OTU2               | 38  | 500  | OTU2               | 38  | 50  |
| Total              | 100 | 1000 |                    | 100 | 100 |

| Standard Tests for Difference |        |        |        |
|-------------------------------|--------|--------|--------|
| P-value                       | chi-2  | Prop   | Fisher |
| Original                      | 0.0290 | 0.0290 | 0.0272 |
| Rarefied                      | 0.1171 | 0.1171 | 0.1169 |

# Does rarefying solve the problem?

---

MINI REVIEW article

Front. Microbiol., 15 November 2017

Sec. Systems Microbiology

Volume 8 - 2017 | <https://doi.org/10.3389/fmicb.2017.02224>

## Microbiome Datasets Are Compositional: And This Is Not Optional

*Gregory B. Gloor<sup>1\*</sup>, Jean M. Macklaim<sup>1</sup>, Vera Pawlowsky-Glahn<sup>2</sup> and Juan J. Egozcue<sup>3</sup>*

<https://doi.org/10.3389/fmicb.2017.02224>

 | Editor's Pick | Microbial Ecology | Research Article | 6 December 2023



Waste not, want not: revisiting the analysis that called into question the practice of rarefaction

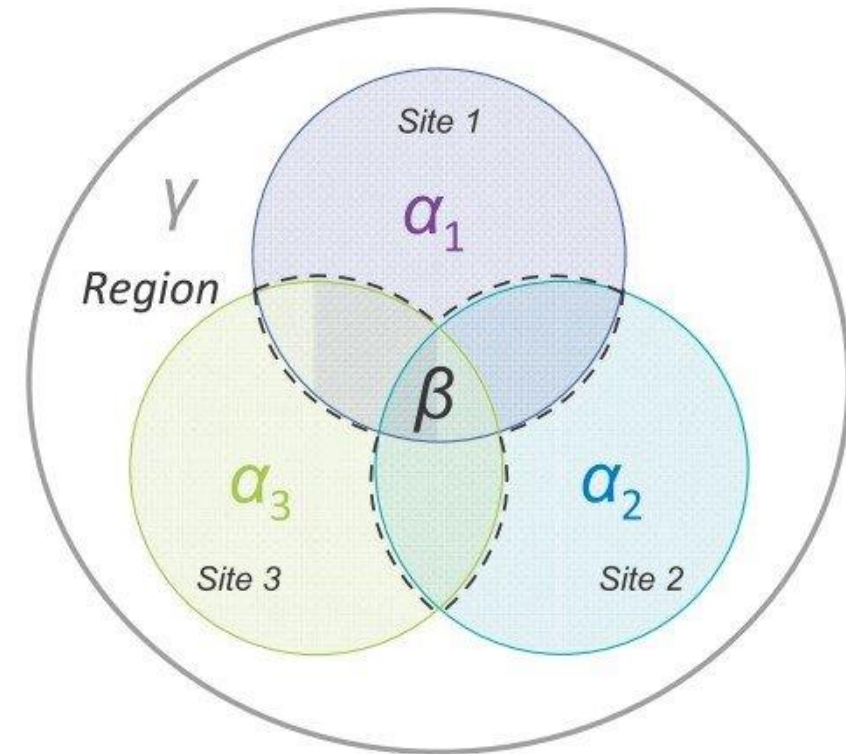
Author: [Patrick D. Schloss](#)   | [AUTHORS INFO & AFFILIATIONS](#)

<https://doi.org/10.1128/msphere.00355-23>

# Diversity measures

---

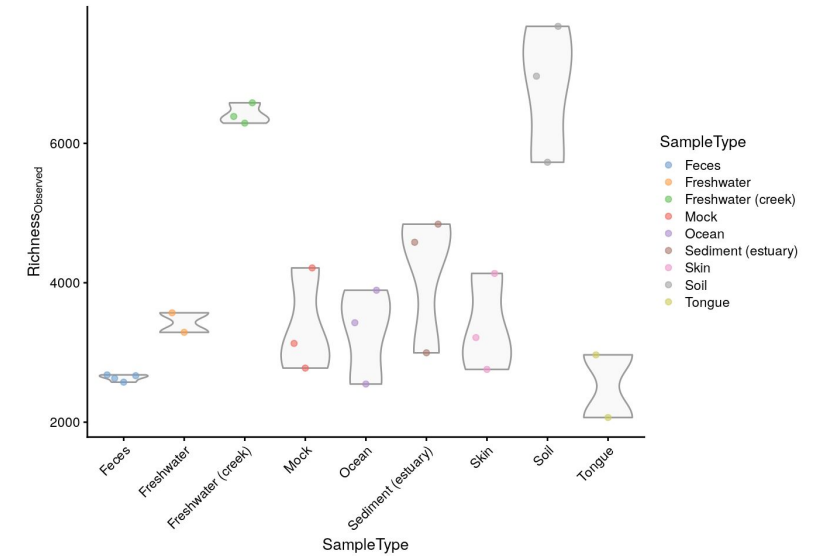
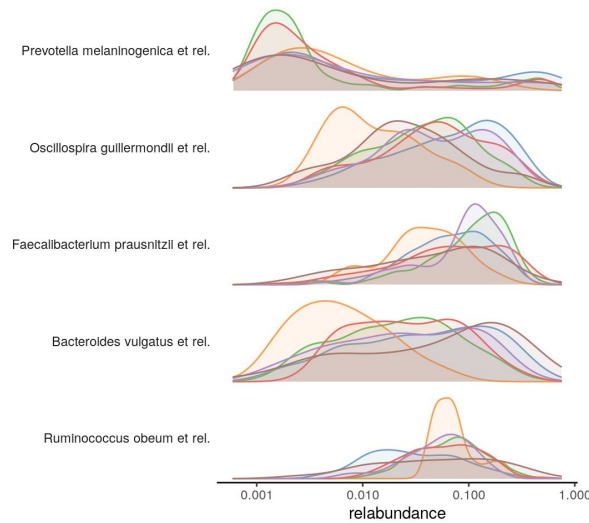
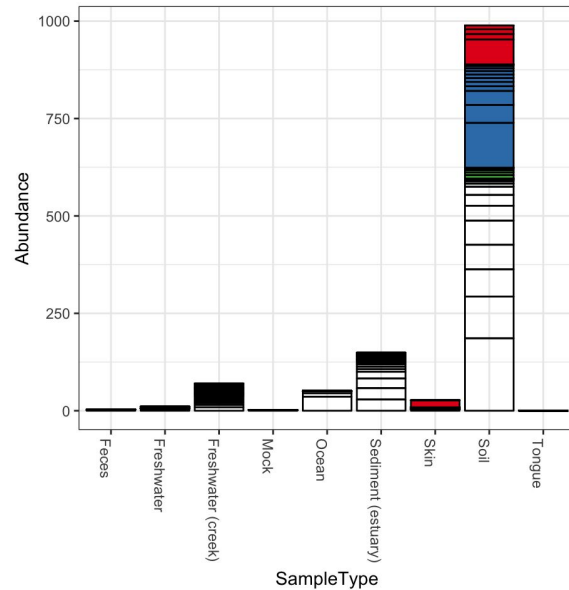
- Alpha diversity
  - Richness (number of different species),
  - Evenness (whether the species are present in similar even abundances),
  - Shannon index
  - Simpson Diversity
  - Chao1 (predicts the total number of species in a sample based on the rarefaction curve)
- Beta diversity
  - Bray-Curtis index, Jaccard index, Aitchison distance, Unifrac distances
- Gamma diversity
  - Estimates diversity within a region



*Credit to Jurgens, 2018.*

# Exploration of microbiome data

Graphical summary, abundance bar, richnees plot.



## ● Abundance bar

GlobalPatterns data, Phyloseq package

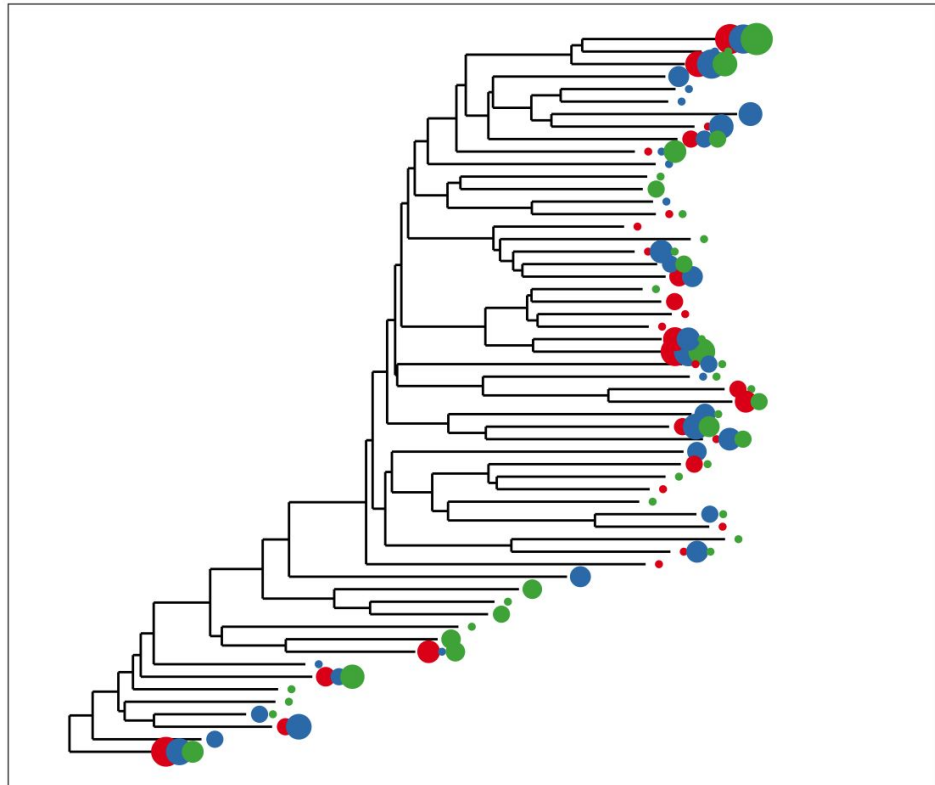
## » Richnees plot

GlobalPatterns data, mia package

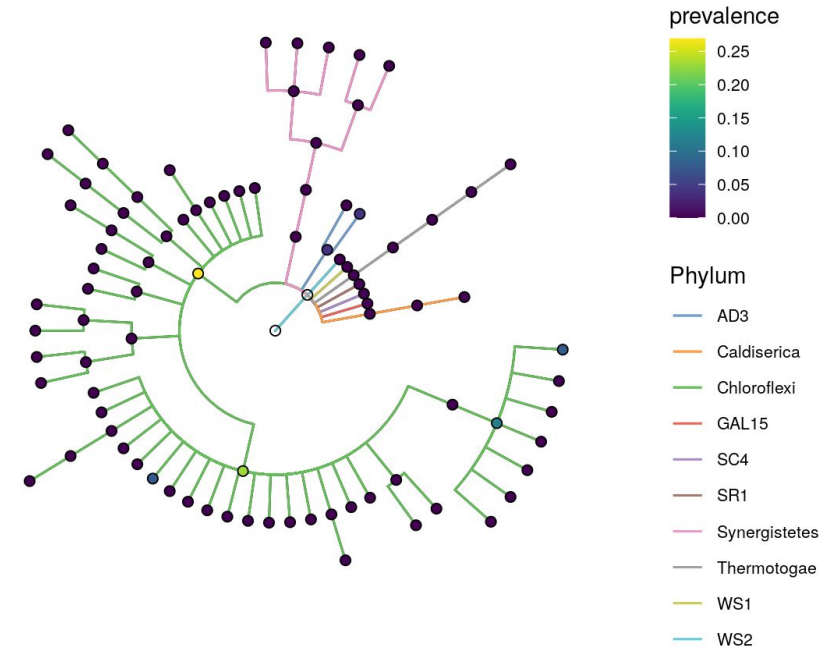
## » Relative abundance

# Exploration of microbiome data

*Phylogenetic trees.*



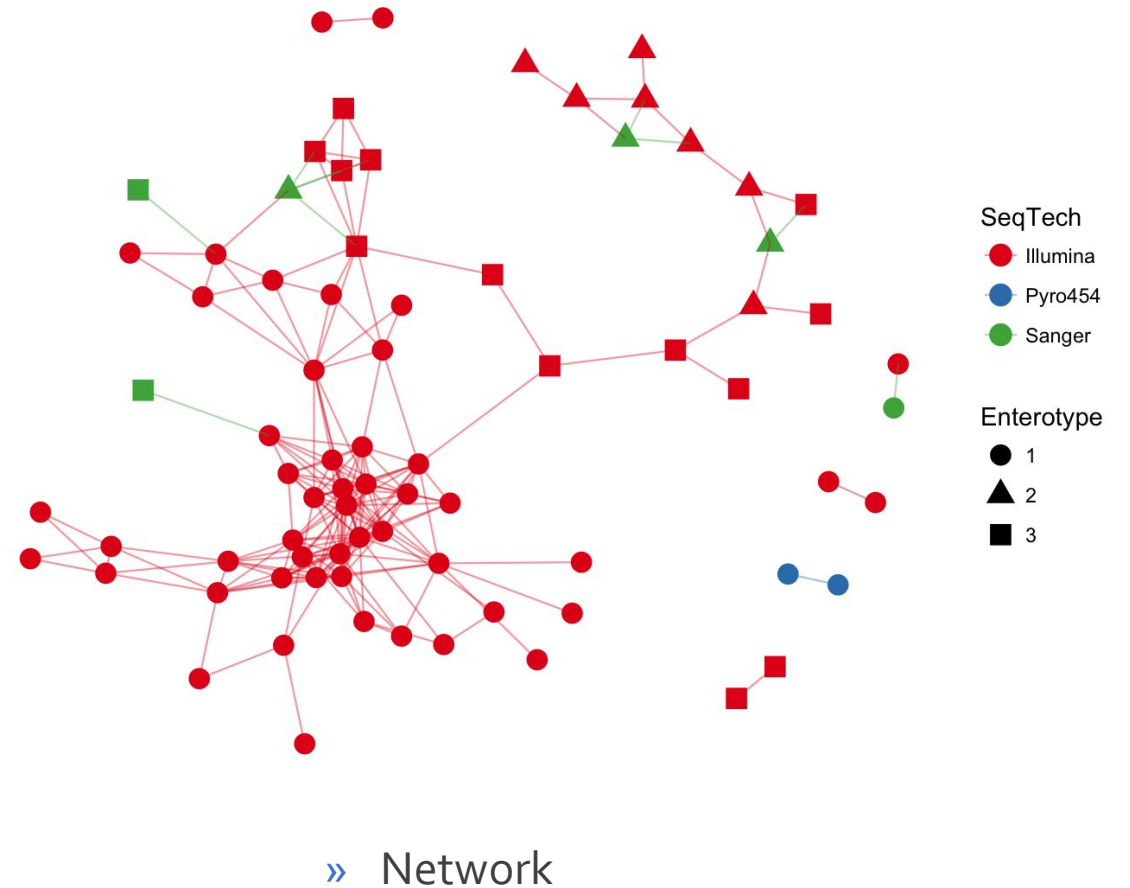
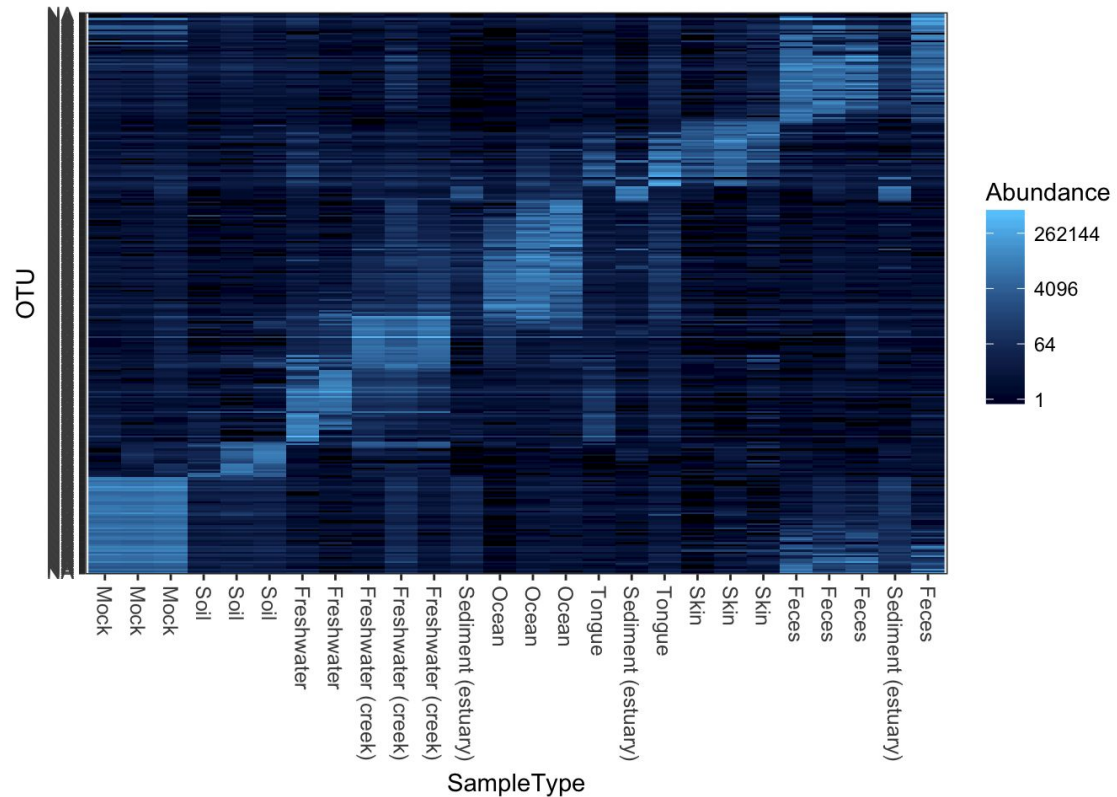
- Esophagus dataset tree, [Phyloseq](#)



» Globalpatterns dataset tree, prevalence of top phyla

# Exploration of microbiome data

*Heatmap and networks.*



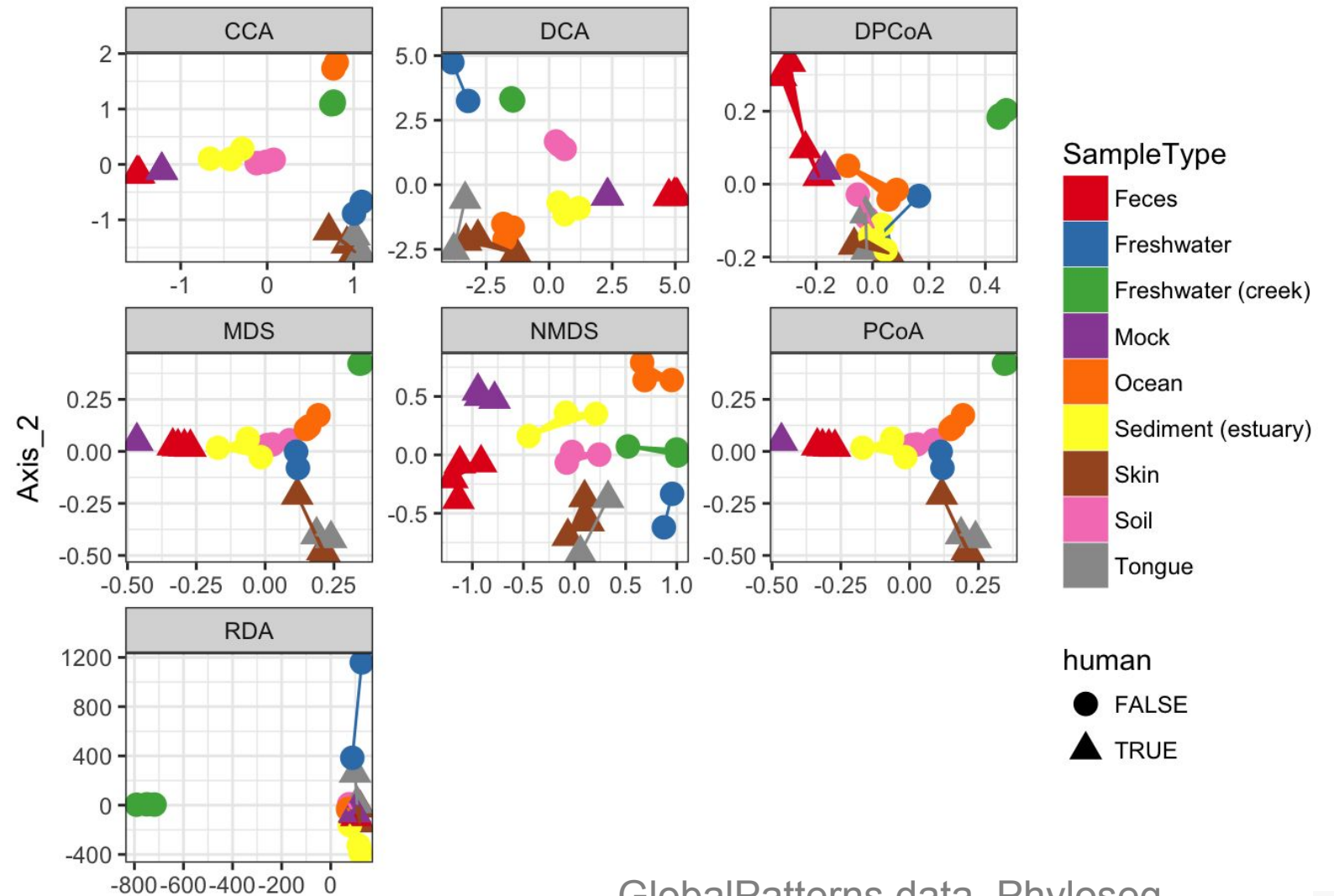
● Heatmap

GlobalPatterns data, [Phyloseq](#)

# Exploration of microbiome data

## Ordination methods

- Canonical Correspondence Analysis (CCA)
- Principal Cordinate Analysis (PCoA)
- Non-metric Multidimensional Scaling (NMDS)
- Redundancy Analysis (RDA)

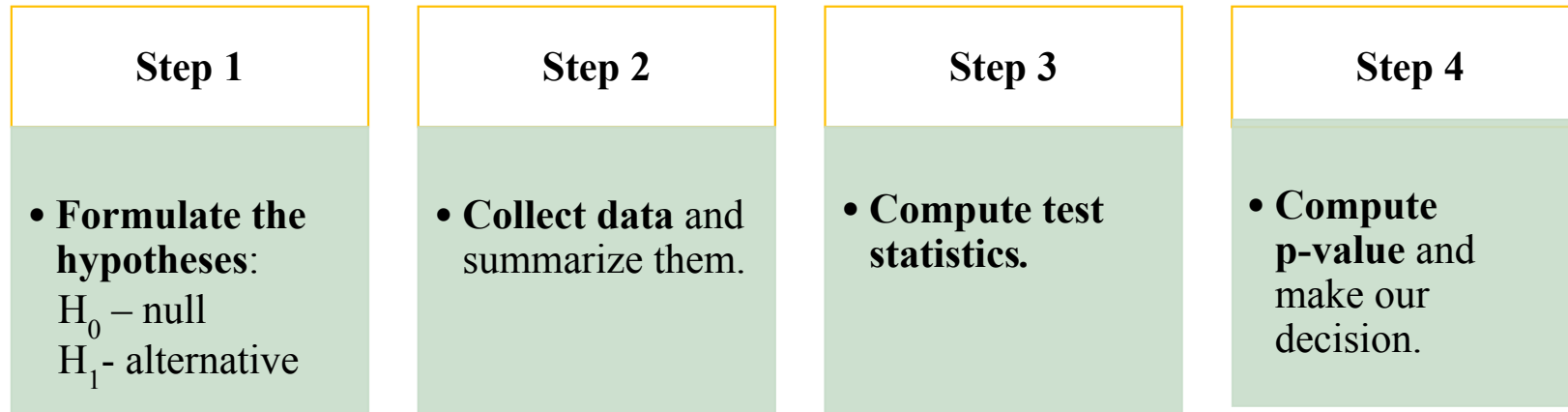


GlobalPatterns data, Phyloseq

## A refresher on hypotheses testing

---

*Statistical hypothesis testing is defined as assessing evidence provided by the data against the null hypothesis.*



1. The null hypothesis ( $H_0$ ) is 'no real difference'.
2. A test statistic (t, U, F, ...) is just a standardized distance between what you observed and what  $H_0$  predicts, the p-value is that distance translated into a probability.

## Two ways a test can be wrong

---

*Type I/II error are the  $\alpha$  and the  $\beta$  from sample-size calculation.*

**$H_0$  true, don't reject**

Correct, true negative

**$H_0$  true, reject**

Type I error ( $\alpha$ ), false positive

**$H_0$  false, don't reject**

Type II error ( $\beta$ ), false negative

**$H_0$  false, reject**

Correct, power =  $1 - \beta$

# Microbiome data usually fails normality and equal-variance checks

---

Compositional

Sparse

Overdispersed

Non-normal

## Parametric tests

*Assume normal distribution, equal variance across groups*

1. Paired / unpaired t-test (2 groups)
2. One-way ANOVA (3+ groups)

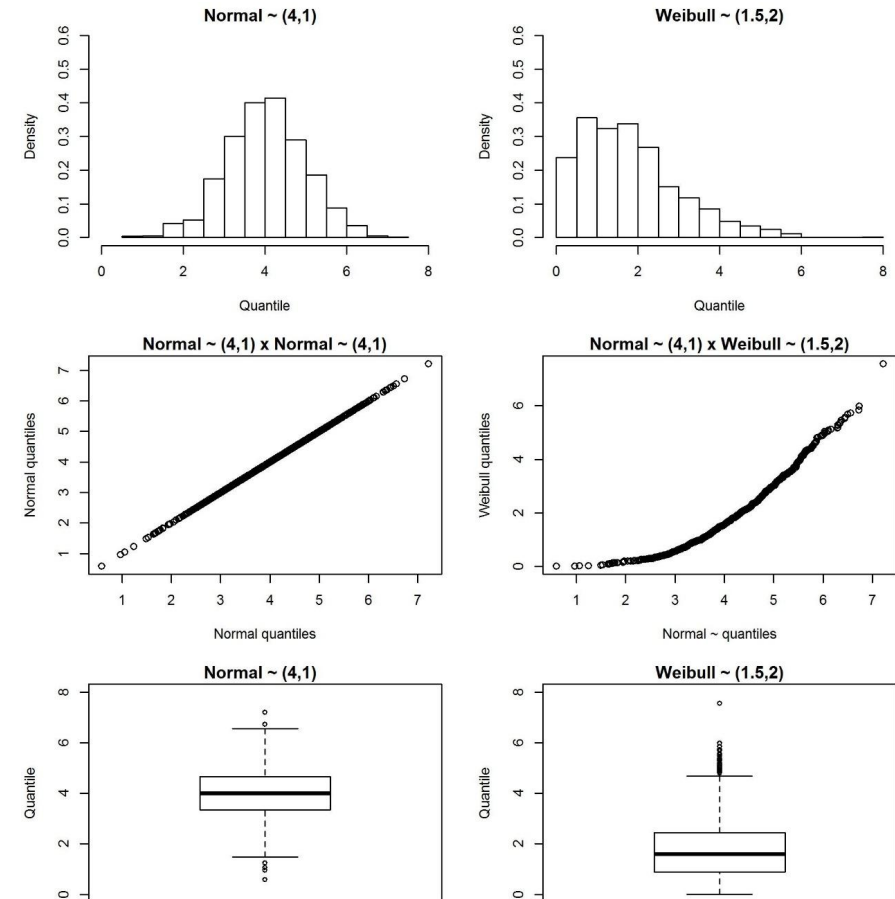
## Non-parametric tests

*Assume nothing about the shape of the distribution*

1. Wilcoxon signed-rank (paired, 2 groups)
2. Mann-Whitney U / Wilcoxon rank-sum (unpaired, 2 groups)
3. Kruskal-Wallis (3+ groups)

# Checking assumptions, and letting your design pick the test

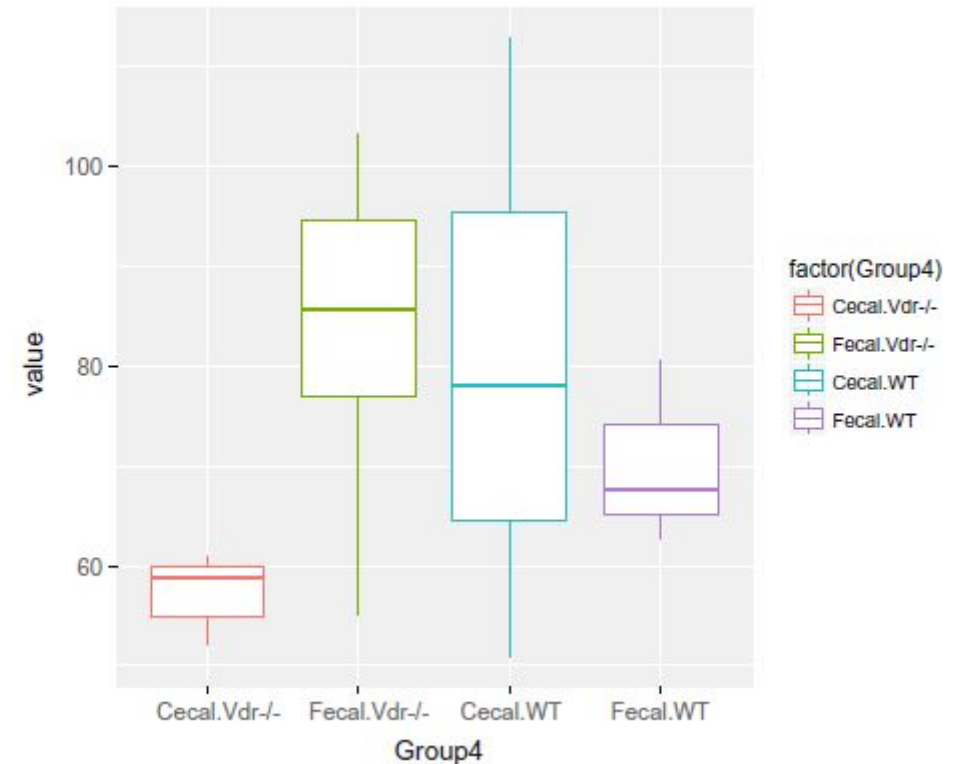
1. Shapiro-Wilk tests normality directly; a quick histogram or Q-Q plot does the same job by eye.
2. Levene's test checks equal variance across groups; unequal variance alone can distort a standard t-test's p-value even when means genuinely differ.



## Comparing diversity indices between groups

*Shannon, Simpson, and richness are single numbers per sample, treat them like any continuous variable, carefully.*

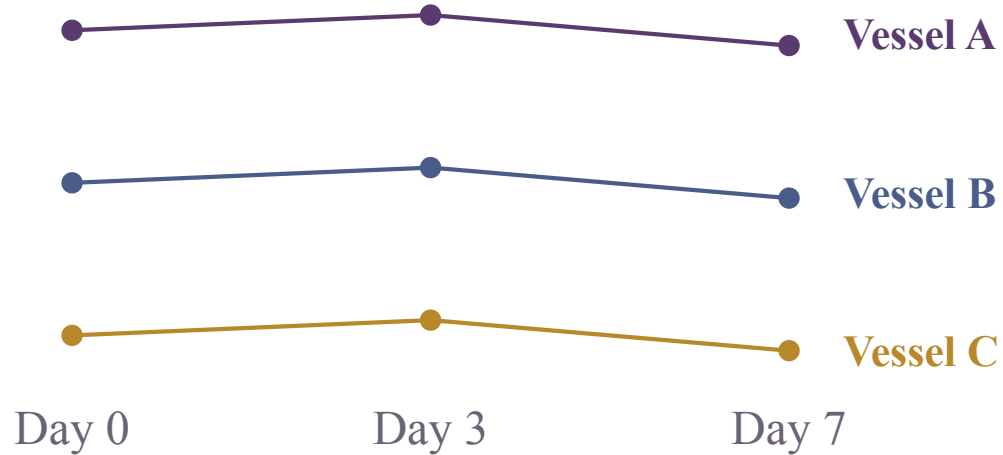
1. Alpha diversity indices (Shannon, Simpson, observed richness) collapse a whole community into one number per sample.
2. Independent samples, non-normal or skewed → Wilcoxon rank-sum (2 groups) or Kruskal-Wallis (3+ groups); no distributional assumption required.
3. Report the median difference and a non-parametric effect size alongside the p-value.



-Analysis of Chao 1 alpha diversity measures; Xia et al., 2018. Springer Series in Statistics

## When the same vessel is sampled more than once

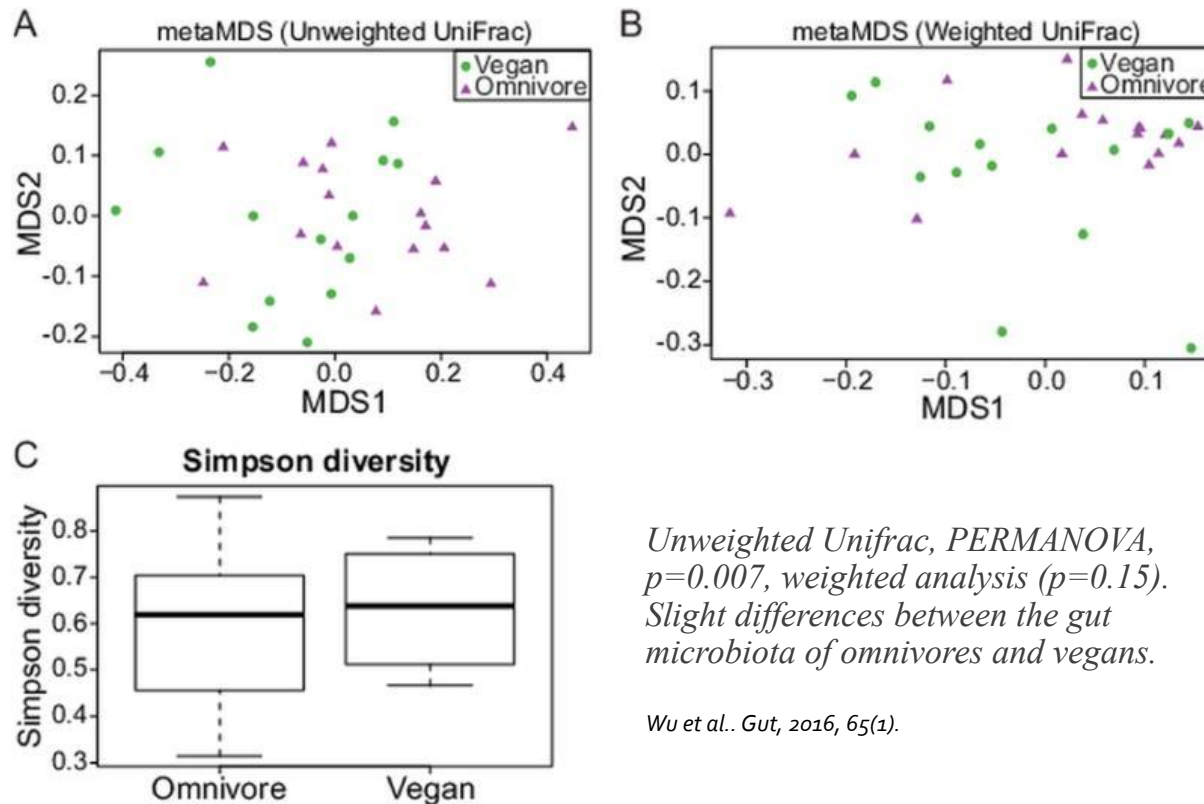
---



1. Treating all 9 points as 9 independent samples overstates your effective sample size and can bias the test.
2. A linear mixed-effects model adds a random intercept per vessel:  $\text{diversity} \sim \text{treatment} + (1 \mid \text{vessel})$ . The vessel term absorbs each vessel's baseline.
3. A repeated-measures ANOVA is a special case for normally-distributed data; for skewed data, use a mixed model on rank-transformed values.

# PERMANOVA tests whole-community differences

*Ordination shows you the pattern. PERMANOVA is what actually tests it.*



*Unweighted Unifrac, PERMANOVA,  $p=0.007$ , weighted analysis ( $p=0.15$ ). Slight differences between the gut microbiota of omnivores and vegans.*

*Wu et al.. Gut, 2016, 65(1).*

1. Distances between samples come from Bray-Curtis, Jaccard, or UniFrac (a full matrix, not one number).
2. PERMANOVA partitions variance in the distance matrix by group, using permutation rather than a theoretical distribution to get a p-value.
3. Flexible to dissimilarity measure; No assumption of multivariate normality; Not sensitive to differences in correlation structure among groups; Can include random effects, interaction terms, and hierarchical structures.

## Which distance metric, for which question

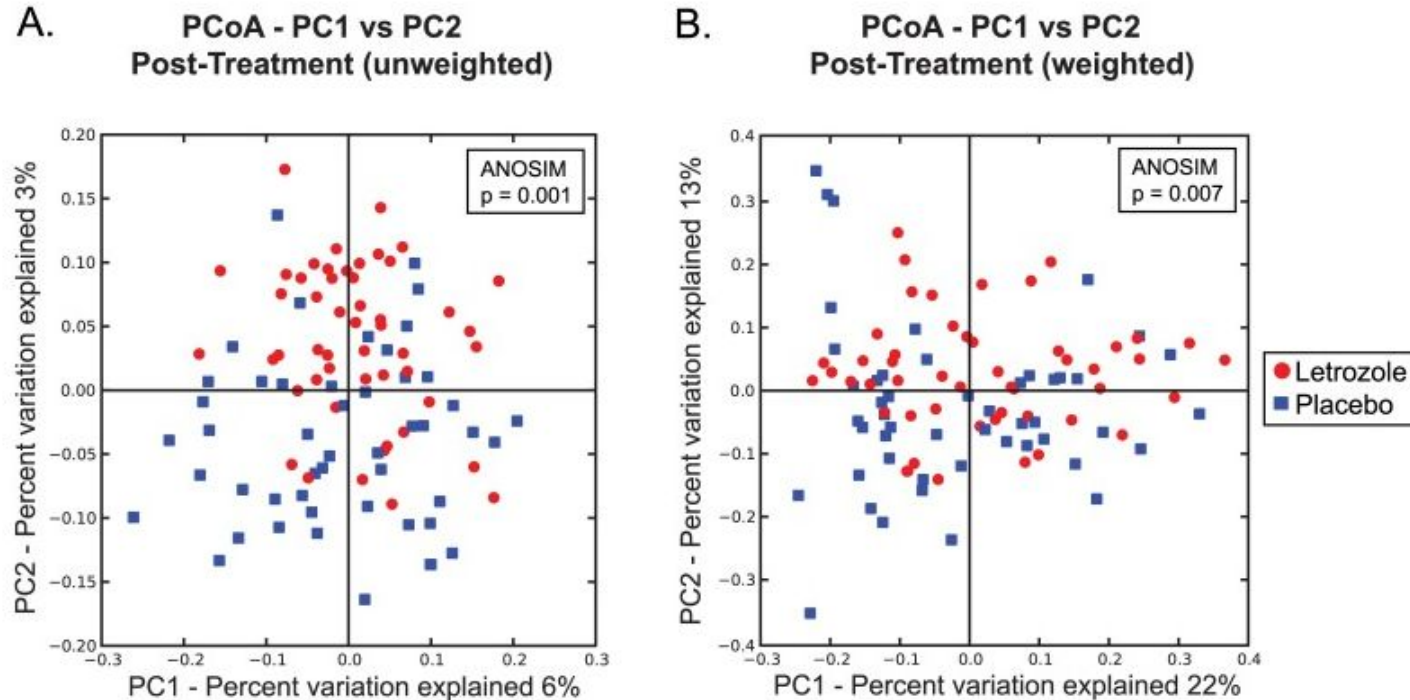
---

| Metric               | Uses abundance?             | Weights rare taxa                    | Good default when...                                      |
|----------------------|-----------------------------|--------------------------------------|-----------------------------------------------------------|
| Jaccard              | No, presence/absence only   | Heavily (every taxon counts equally) | You care which taxa are present or absent, not how much   |
| Bray-Curtis          | Yes, abundance-weighted     | Lightly                              | You care about shifts in the dominant community structure |
| UniFrac (unweighted) | No, presence/absence + tree | Heavily                              | You care which lineages are present, using phylogeny      |
| UniFrac (weighted)   | Yes, abundance + tree       | Lightly                              | You care about abundance-weighted phylogenetic shifts     |

*PERMANOVA doesn't care which distance matrix it receives, only that the choice matches your question.*

# ANOSIM

*Nonparametric test which operates on a ranked dissimilarity matrix.*



The null hypothesis is that the similarities within sites are smaller or equal to the similarities between sites.

*Test the association of microbiome composition between treatments and among time points within treatments using weighted and unweighted UniFrac distances. Kelley et al., 2016. PloS one, 11(1)*

# Compositional data analysis

---

*Don't want to throw away data?*

- Don't rarefy
- Apply CLR (centre log ratio) transformation
  - Balances all features, avoids dominance by high-abundance taxa
  - Requires the addition of “pseudocounts”, zeros are converted to some “small” number.

# Purpose-built methods for compositional data

- 1. Purpose-built methods model the compositional structure directly instead of testing raw relative abundances one taxon at a time, the mechanics are beyond today's scope, but know they exist and why they're preferred.
- 2. The more important choice is using one of them at all, instead of a naive per-taxon test.

*Love, Huber & Anders (2014), Genome Biology — DESeq2;*  
*Robinson, McCarthy & Smyth (2010), Bioinformatics — edgeR;*  
*Fernandes et al. (2014), Microbiome — ALDEx2.*

| Operation               | Standard approach                       | Compositional approach   |
|-------------------------|-----------------------------------------|--------------------------|
| Normalization           | Rarefaction<br>'DESeq'                  | Centered Log Ratio (CLR) |
| Distance                | Bray-Curtis<br>UniFrac<br>Jenson-Shanon | Aitchison                |
| Ordination              | PCoA<br>(Abundance)                     | PCA<br>(Variance)        |
| Multivariate comparison | perMANOVA<br>ANOSIM                     | perMANOVA<br>ANOSIM      |
| Correlation             | Pearson<br>Spearman                     | SparCC<br>SpiecEasi      |
| Differential abundance  | metagenomSeq<br>LEfSe<br>DESeq          | ALDEx2<br>ANCOM          |

# Converting relative abundance to absolute abundance

---

1. Quantify the number of cells or amount of DNA for each sample
  - qPCR
  - Flow cytometry
  - Spike in known number of cells or DNA
2. Multiply each sample by their quantity to obtain absolute numbers and avoid compositionality.

Received: 19 July 2019 | Revised: 15 November 2019 | Accepted: 15 November 2019  
DOI: 10.1002/mbo3.977

ORIGINAL ARTICLE

MicrobiologyOpen WILEY

**Absolute quantitation of microbes using 16S rRNA gene metabarcoding: A rapid normalization of relative abundances by quantitative PCR targeting a 16S rRNA gene spike-in standard**

Olivier Zemb<sup>1</sup> | Caroline S. Achard<sup>2</sup> | Jerome Hamelin<sup>3</sup> | Marie-Léa De Almeida<sup>1</sup> |  
Béatrice Gabinaud<sup>1</sup> | Laurent Cauquil<sup>1</sup> | Lisanne M.G. Verschuren<sup>4,5,6</sup> |  
Jean-Jacques Godon<sup>3</sup>

**One test at  $\alpha = 0.05$  is safe. A thousand tests is not.**

---

|                   | Family wise error (FWER)                                                  | False discovery rate (FDR)                                                           |
|-------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Controls</b>   | P(at least one false positive), anywhere                                  | Expected proportion of false positives among significant results                     |
| <b>Correction</b> | Bonferroni (or Holm, a less conservative stepwise version)                | Benjamini-Hochberg                                                                   |
| <b>Best when</b>  | One false positive would be a serious problem (e.g. a confirmatory claim) | You expect many true effects and can tolerate a small, known fraction of false leads |
| <b>Trade-off</b>  | Conservative, real effects get missed as m grows                          | Sacrifices some FWER control for much greater power                                  |

*Exploratory 16S / shotgun screens of hundreds of taxa are almost always FDR territory. Benjamini-Hochberg dominates this field.*

# Bonferroni controls FWER; Benjamini-Hochberg controls FDR

---

## BONFERRONI · FWER · CONSERVATIVE

$$\alpha_{adjusted} = \alpha / m$$

Guarantees  $\text{FWER} \leq \alpha$ . Simple, but extremely conservative when  $m$  is large, real effects get missed.

## BENJAMINI-HOCHBERG · FDR · THE OMICS STANDARD

$$\text{Reject } p_{(1)} \dots p_{(k)}, k = \max i : p_{(i)} \leq (i/m) \cdot \alpha$$

Controls expected FDR at  $\alpha$ . More power than Bonferroni, the standard choice for omics-scale testing.

*Benjamini & Hochberg (1995), J R Stat Soc B, the original FDR paper, one of the most cited papers in statistics.*

## Applying Benjamini-Hochberg to 10 taxa

---

*Rank p-values smallest to largest, compare each to its own threshold, and find the last one that still passes.*

| Rank (i) | Taxon    | p-value | BH threshold ( $i/10 \times 0.05$ ) | Result          |
|----------|----------|---------|-------------------------------------|-----------------|
| 1        | Taxon 1  | 0.001   | 0.005                               | Significant     |
| 2        | Taxon 2  | 0.004   | 0.010                               | Significant     |
| 3        | Taxon 3  | 0.006   | 0.015                               | Significant     |
| 4        | Taxon 4  | 0.011   | 0.020                               | Significant     |
| 5        | Taxon 5  | 0.019   | 0.025                               | Significant     |
| 6        | Taxon 6  | 0.028   | 0.030                               | Significant     |
| 7        | Taxon 7  | 0.045   | 0.035                               | Not significant |
| 8        | Taxon 8  | 0.060   | 0.040                               | Not significant |
| 9        | Taxon 9  | 0.130   | 0.045                               | Not significant |
| 10       | Taxon 10 | 0.290   | 0.050                               | Not significant |

*6 of 10 taxa remain significant after BH correction, vs. 7 that would pass an uncorrected  $p < 0.05$  cutoff, taxon 7 ( $p = 0.045$ ) looks significant alone, but fails once you account for testing 10 hypotheses.*

# Statistically significant is not the same as biologically important

---

1. A p-value says nothing about how big or meaningful the difference is.
2. With enough samples, even a practically irrelevant difference becomes 'statistically significant'.
3. Effect size answers the question a p-value can't: how large is the difference?
4. Always report both the p-value and effect size.

## Large effect + Significant

Real and important

## Small effect + Significant

Statistically, not practically important

## Large effect + Not significant

Likely underpowered

## Small effect + Not significant

Nothing here

## The exact same tiny effect, at different sample sizes

---

*Fixed throughout: a 0.3-percentage-point difference in relative abundance,  $d = 0.15$  (most people would call this irrelevant).*

| n per group | p-value | Significant at $\alpha = 0.05$ ? |
|-------------|---------|----------------------------------|
| 10          | 0.74    | No                               |
| 50          | 0.45    | No                               |
| 100         | 0.29    | No                               |
| 500         | 0.018   | Yes                              |
| 1,000       | 0.0008  | Yes                              |

Push n to 5,000 and the same effect reaches  $p < 1 \times 10^{-13}$

## Effect size measures

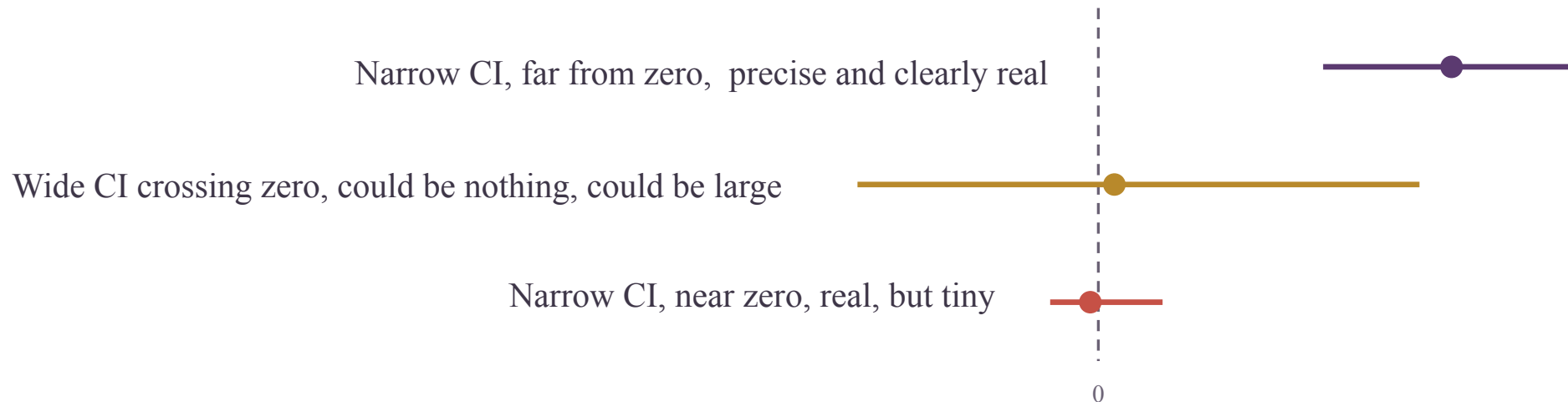
---

| Measure                         | Use it for                                                                   | Rough interpretation                                      |
|---------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------|
| Cohen's d / Hedges' g           | Comparing means of a continuous variable (diversity index, metabolite level) | 0.2 small, 0.5 medium, 0.8 large                          |
| Log2 fold change                | Per-taxon differential abundance                                             | LFC = 1 → taxon is 2× more abundant                       |
| PERMANOVA R <sup>2</sup>        | Community-level (beta diversity) comparisons                                 | Share of total community variance explained by the factor |
| Cliff's delta / rank-biserial r | Non-parametric comparisons (Wilcoxon / Mann-Whitney)                         | 0.11 small, 0.28 medium, 0.43 large                       |
| Odds ratio / risk ratio         | Presence/absence or binary outcomes                                          | 1 = no association; different from 1 = significant        |

## A confidence interval does more work than a p-value

---

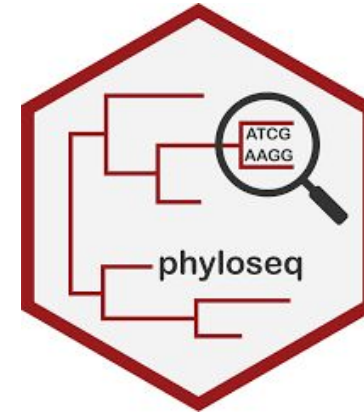
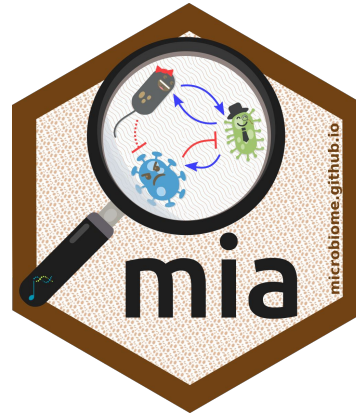
1. A 95% CI on your effect size shows both its magnitude and your uncertainty about it, a p-value collapses both into a single yes/no.
2. A narrow CI far from zero and a wide CI that barely excludes zero can share the same p-value.
3. For multiple comparisons, adjust the CIs too, an unadjusted CI is just as optimistic as an unadjusted p-value.



# Software to implement the analysis

---

1. mia
2. phyloseq
3. Vegan
4. biom
5. DESeq
6. DESeq2
7. metagenomeSeq
8. ANCOM
9. ALDEx and ALDEx2





1. Check normality and equal variance before defaulting to a t-test or ANOVA, microbiome data usually fails both.
2. Alpha diversity: rank-based tests or mixed models for repeated measures. Beta diversity: PERMANOVA on the distance matrix, ordination shows the pattern, it doesn't test it.
3. Differential abundance needs compositional-aware tools (ALDEx2, ANCOM-BC), not per-taxon t-tests, and not rarefying.
4. Correct for multiple comparisons, and know whether you're controlling FWER (Bonferroni) or FDR (Benjamini-Hochberg).
5. A p-value and an effect size answer different questions, report both, every time, even after correction.

## Resources/Courses

---

- **Fundamentals of Biostatistics with R**  
<https://github.com/e-ibrahimi/Fundamentals-of-Biostatistics-with-R->
- **Generalized linear models with R**  
<https://github.com/e-ibrahimi/Generalized-Linear-Models-with-R>
- **Beginner Microbiome Analysis**  
[https://bioinformaticsdotca.github.io/BMB\\_2025/](https://bioinformaticsdotca.github.io/BMB_2025/)
- **Advance Microbiome Analysis**  
[https://bioinformaticsdotca.github.io/AMB\\_2025/](https://bioinformaticsdotca.github.io/AMB_2025/)
- **Orchestrating microbiome analysis with Bioconductor**  
<https://microbiome.github.io/OMA/resources.html#data-containers-1>
- Ibrahim, E., Woford, B.N. **Ten essential tips for robust statistics in cell biology.** *Nat Cell Biol* **27**, 1884–1886 (2025). <https://doi.org/10.1038/s41556-025-01801-y>

# Questions?

---